1. Suppose that in 2020 MOE was given a sharp increase in funding for its two flagship universities NUS and NTU. MOE decided that the funding would go to NTU, mostly to hire more teachers to reduce class sizes. To evaluate the effectiveness of the increased funding, MOE collected random samples of the students in 2019, 2021, and 2022 at both universities. Suppose the measure of student performance is a standardized test score given to all undergraduate students in the country.

   (a) Let *score* be the test score. Without controlling for other factors, set up a difference-in-differences estimation by using a linear regression model that allows you to determine whether the additional funding led to improved test scores for 2021 and 2022, respectively. Explain the interpretation of the parameters in your model. [Hint: you need to find the DID effects for 2021 and 2022 separately.]

   (b) Give a graphical illustration of your model. What is the key assumption of your model? Explain.

2. Consider the following linear regression model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + u_i,\ i = 1, ..., n. \tag{1}$$

   We want to test the hypothesis $H_0 : \beta_1 = \beta_2$ vs. $H_1 : \beta_1 \neq \beta_2$. This null hypothesis imposes a single restriction ($q = 1$) but on two coefficients (β_1 and β_2).

   To test the above hypothesis, we substract and add $\beta_2 X_{1i}$ and obtain

$$Y_i = \beta_0 + (\beta_1 - \beta_2)X_{1i} + \beta_2(X_{1i} + X_{2i}) + u_i, \tag{2}$$

   which is equivalent to the following model:

$$Y_i = \beta_0 + \gamma_1 X_{1i} + \beta_2 W_i + u_i, \tag{3}$$

   where $\gamma_1 = \beta_1 - \beta_2$ and $W_i = X_{1i} + X_{2i}$. Discuss how we can test $H_0 : \beta_1 = \beta_2$ vs. $H_1 : \beta_1 \neq \beta_2$ by using a standard t-test and the model in (3).